

SmartBin-1

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Project ID: 03

Project Title: SmartBin-1: Automated IoT Waste-Level Monitoring & Alert System

Team Size: Team (2 Students)

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Project Category: Electronics/ Embedded

Project Type: Prototype Development

Current Development Stage: Testing Complete

Brief Project Description: SmartBin-1 is an IoT-enabled waste bin monitoring node for campus environments. It uses an HC-SR04 ultrasonic sensor and an ESP32 microcontroller to continuously measure how full a bin is, converts that distance reading into a fill percentage, and reports it through on-bin LEDs, an alarm buzzer, and a live web dashboard. The ESP32 hosts its own Wi-Fi Access Point, so the dashboard works on any phone or laptop with no internet connection or external router.

Engineering Problem Statement: Municipal and institutional waste collection currently relies on fixed inspection schedules rather than actual fill data. This causes overflowing bins between rounds, wasted collection trips to bins that are still empty, and no usable data for planning collection routes.

Motivation: Campus bins in high-footfall areas (canteens, hostels, libraries) fill unevenly through the day and are often checked too late or too early. A low-cost sensor node that reports its own state removes the guesswork and lets cleaning staff respond to actual conditions instead of a fixed timetable.

Existing System/Technology: Manual visual inspection by housekeeping/sanitation staff on a fixed round; in some commercial deployments, basic fill-sensors tied to cloud dashboards requiring continuous internet and subscription infrastructure.

Limitations of Existing System: Manual rounds are labour-intensive, reactive rather than predictive, and do not scale to many bins. Existing commercial IoT bins typically require persistent internet/cloud connectivity and recurring subscription costs, which makes them expensive to pilot at small scale (e.g. a single campus building).

Project Objectives:

1. Measure container fill level continuously and without physical contact.
2. Convert raw ultrasonic distance into a calibrated 0–100% fill percentage.
3. Provide immediate on-site signalling: green (OK), yellow (half full), red + buzzer (full).
4. Expose live readings through a web dashboard that works entirely offline over a local Wi-Fi hotspot.
5. Keep hardware cost low enough for multi-bin deployment.

Expected Engineering Outcome: A functioning low-cost sensor node (prototype cost INR 870 in components) that measures fill level with a worst-case error of about $\pm 2.88\%$, signals status locally, and serves a real-time dashboard without needing internet or cloud infrastructure — validated through bench testing at three fill states (OK, half full, full).

Research Gap Identified: Most existing smart-bin literature and commercial products assume continuous internet/cloud connectivity for telemetry, which raises cost and limits deployment to connected sites. There is limited documented work on a fully standalone (offline Access Point) fill-monitoring node that still provides a real-time dashboard without any external network dependency.

How the Project Addresses the Research Gap: SmartBin-1 removes the internet/cloud dependency entirely by having the ESP32 host its own Wi-Fi Access Point and serve the dashboard directly over HTTP, so a single low-cost node can be piloted at any site with no network infrastructure, router, or recurring connectivity cost.

Literature Review: Several studies describe ultrasonic-sensor-based smart bins. Thangaraj et al. (2020) built a NodeMCU-based bin with an ultrasonic sensor and buzzer that pushes a fill alert to an IoT platform once a threshold is crossed. A 2026 Zenodo-published system extends this with GSM and GPS, sending an authority alert with the bin's location once fill exceeds 98%, but depends on cellular connectivity for every reading. The i-SMARTBiN project (Malaysian Institute of Road Safety Research) adds an

IR sensor and automatic lid control alongside ultrasonic level sensing. More recently, an arXiv preprint, CleanCity-BinSense (2026), proposes a configurable, geometry-independent fill-percentage formula and a 3D-printed sensor enclosure, paired with nearest-neighbour route optimisation across many bins.

Broader surveys confirm the trend: Shah et al. (2021, IJEECS) reviewed 16 IoT waste-monitoring papers and found most rely on Arduino Uno with varying sensor/communication choices; Anagnostopoulos et al. (2017, IEEE Trans. Sustainable Computing) surveyed ICT-enabled waste management broadly; and Sanchez Portugal & Salazar (2025) performed a PRISMA-based review of 23 IoT waste-management studies, noting that initial cost and required network infrastructure remain common barriers to deployment, especially in lower-resource settings.

Across this literature, almost all systems assume continuous connectivity — Wi-Fi to a router, GSM, or a cloud/LoRaWAN backend — to make readings visible remotely. SmartBin-1 differs by hosting its own Wi-Fi Access Point and dashboard directly on the ESP32, so live monitoring works with zero external network infrastructure, addressing the cost/infrastructure barrier these reviews identify.

Existing Products/Technologies Identified: Cloud-connected commercial smart bins (e.g. compactor/sensor combinations used in some smart-city pilots) that report fill level via cellular or Wi-Fi to a central server; academic prototypes using ultrasonic or infrared sensors with Arduino/ESP8266/ESP32 reporting over MQTT or HTTP to cloud dashboards.

Patent Search Conducted? No

Patent Databases Used: NA

Relevant Patents Identified: NA

Comparison with Existing Patents: NA

Main Innovation of the Project: A fully standalone fill-monitoring node: ultrasonic sensing, threshold-based alerting, and a live web dashboard all served locally from the ESP32's own Access Point, with zero dependency on internet, cloud services, or external servers.

Novel Technical Features:

1. Local Wi-Fi Access Point hosting the dashboard directly from the microcontroller (no router/cloud).
2. Combined local (LED + buzzer) and remote (dashboard) alerting from the same threshold logic.
3. Safe 5V-to-3.3V signal conditioning for the ultrasonic echo pin using a simple resistive divider, keeping the bill of materials minimal.
4. Browser-side cleaning-event detection (auto-logs when a full bin transitions back to OK).

Difference from Existing Solutions: Existing IoT bin monitors generally require a network/cloud backend to view data remotely; SmartBin-1 serves the same real-time view locally, making it deployable in locations with no internet access (e.g. temporary setups, remote yards, pilot sites).

Technical Advantages / Expected Improvements: Lower deployment cost (no subscription/cloud fees), faster pilot setup (no network provisioning), and resilience to internet outages since the dashboard runs entirely on device.

Potential Patent/IP? No

Design Methodology: Iterative hardware-software co-design:

- (1) define fill-percentage formula and alert thresholds
- (2) select and wire sensor/output components on breadboard,
- (3) implement firmware for ranging thresholding and the embedded dashboard
- (4) bench-calibrate against a fixed bin depth
- (5) validate across the three alert states.

Design Requirements/Specifications:

1. Non-contact fill measurement, update cycle ≤ 1 second.
2. Three distinct alert states with both local (LED/buzzer) and remote (dashboard) indication.
3. Operate without internet or external router.
4. Component cost low enough for multi-unit deployment.
5. Safe voltage levels at all microcontroller inputs.

Engineering Assumptions:

1. Bin is rigid with a flat waste surface for ranging purposes.
2. Sensor mounted at a fixed, known depth (14.4 cm) from the empty-bin floor.
3. Ambient temperature near 20°C, giving speed of sound ≈ 343 m/s.

4. Single sensor view represents the general fill state (no multi-point averaging).

Major Design Parameters:

1. Calibrated empty-bin depth: 14.4 cm
2. Full-alert threshold distance: 4.0 cm
3. Alert bands: OK > 9.2 cm, Half-full 6.08–9.2 cm, Full $\leq$ 6.08 cm
4. Sensing cycle: 200 ms; dashboard refresh: 500 ms
5. Worst-case measurement error: $\pm 2.88\%$

Design Calculations Report: PDF**CAD Software Used:** Others**CAD Model/Assembly Files:** File / PDF

Bill of Materials: The SmartBin-1 prototype is built around an ESP32 DevKit V1 (30-pin, ESP-WROOM-32) with its USB cable, the main controller that runs the Wi-Fi hotspot and web server (₹529). Distance sensing is done by one HC-SR04 ultrasonic sensor (₹75). The local alerts use one 3 mm green LED (₹1.60), one 3 mm yellow LED (₹1.60), one 5 mm red LED (₹2) and one 5 V active buzzer module (₹29). The circuit needs three 220 Ω resistors for LED current limiting and three 1 k Ω resistors, which form the 1 k Ω /2 k Ω voltage divider on the ECHO line. All six are 1/4 W carbon film and together cost ₹6. Two solderless breadboards (₹140) hold the circuit, and it is wired with 7 female-to-female jumper wires (₹8.75) and 14 male-to-male jumper wires (₹17.50), all 20 cm long. A plastic dustbin container (₹50) serves as the bin, and double-sided tape (₹10) mounts the sensor and buzzer. In total items, costing ₹870.45.

Prototype Description: A breadboard-built node with the ESP32 at the centre, HC-SR04 mounted facing down into the test container, three LEDs and a buzzer wired to GPIOs with current-limiting resistors, and the ECHO line passed through a 1 k Ω & 2 k Ω voltage divider before reaching GPIO18. SmartBin-1 is an ESP32-based smart bin. An HC-SR04 ultrasonic sensor under the jar lid measures the distance to the waste. The ESP32 converts it to a fill percentage (14.4 cm = empty, 4.0 cm = full). Green, yellow and red LEDs and a buzzer show the level, and a live dashboard is served over the board's own Wi-Fi hotspot (192.168.4.1) with no internet.

Fabrication Method and Materials: No custom fabrication - assembled on a standard solderless breadboard using off-the-shelf electronic components and jumper wires, firmware written in C++ (Arduino core for ESP32).

Experimental Setup and Test Procedure: Material was placed inside the calibrated container at three different heights to produce distances of 10.4 cm, 7.7 cm and 5.8 cm

from the sensor, corresponding to the OK, half-full and full bands. At each height, the dashboard reading (distance, fill %, status, active LED, buzzer state) was recorded and compared against the value calculated directly from the fill formula.

Test Parameters and Instruments: Parameters: sensor-to-surface distance (cm), computed fill percentage (%), LED/buzzer state, dashboard status label. Instruments: the HC-SR04 sensor itself (no external reference rangefinder used); a smartphone connected to the SmartBin-WiFi Access Point to view the dashboard; Arduino IDE Serial Monitor at 115200 baud for raw readings.

Processed Data (Source Codes): Source File

Prototype Photographs: PDF of Photos

Demonstration Video: Video

Key Performance Parameters: Fill-percentage accuracy ($\pm$ error vs. sensor's nominal ± 0.3 cm precision), response latency (200 ms sensing cycle, 500 ms dashboard refresh), alert-threshold correctness, Wi-Fi AP range (≈ 15 – 30 m line of sight).

Major Results: All three alert states were correctly triggered and matched expected LED colour, buzzer state, and dashboard status: 10.4 cm $\rightarrow$ 39% (OK, green); 7.7 cm $\rightarrow$ 64% (Half full, yellow); 5.8 cm $\rightarrow$ 83% (Full, red + buzzer). Reported values matched the calculated formula output within rounding.

Key Engineering Findings: The 1 k Ω /2 k Ω resistive divider reliably protects the ESP32 GPIO while preserving usable ECHO timing accuracy; the system maintains consistent <1 second response from physical fill change to dashboard update; worst-case theoretical error from sensor tolerance is $\pm 2.88\%$ over the 10.4 cm working span.

Project Limitations: Single-point measurement can misread uneven waste piles; soft or steeply angled materials can scatter the ultrasonic pulse and cause a missed echo (read as empty); Wi-Fi AP range and client count are limited by the ESP32's built-in antenna; the cleaning-history log is stored in the browser session, not on the device, so it resets on page reload.

Future Improvements/Scopes: Replace the local AP with Lora WAN or NB-IoT for multi-kilometre, cloud-connected deployments; add solar charging with a Li-ion cell and deep-sleep firmware for outdoor/unpowered sites; aggregate data across many nodes to auto-generate collection routes ranked by fill priority.

Research Publication Potential: Yes

Proposed Research Paper Title: SmartBin-1: A Standalone, Internet-Independent ESP32 System for Real-Time Waste-Level Monitoring

Proposed Invention Title: Standalone Access-Point-Based Fill-Level Monitoring and Alert System for Waste Containers

Technical Problem Solved: Real-time, low-cost waste-fill monitoring without dependency on internet connectivity or cloud infrastructure.

Core Technical Innovation: On-device hosting of both the sensing logic and the live dashboard via a self-created Wi-Fi Access Point, eliminating external network/cloud requirements.

Novel Features: Offline dashboard hosting; combined local+remote three-tier alerting from one threshold engine; minimal-component safe signal conditioning for the echo line.

Technical Advantages: Lower cost per unit, faster deployment (no network setup), functions in locations with no internet access, simple to replicate and maintain.

Potential Applications: Campus waste bins, hospital ward/OPD bins, transit-hub bins, and commercial/mall bins.

Student 1 Contribution: Hardware design, circuit wiring, firmware development, dashboard UI, testing and calibration, report writing.

Student 2 Contribution: Component sourcing, breadboard assembly support, test-data recording, report review.

Final Declaration

I/We declare that the submitted information, data, results, references, files and contribution details are accurate and represent our project work. We understand that

publication and patent/IP protection are subject to appropriate academic, editorial, institutional and legal evaluation.